

HTTP 5310 – Capstone Requirements Document

Project Title: HealthCare Plus

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Program: Web Development – Humber College

Course: HTTP 5310 Capstone

1. Introduction and Rationale

HealthCare Plus is a booking web application designed to simplify how patients book medical appointments while providing doctors and administrators with efficient tools to manage schedules, availability, and content.

Many healthcare clinics still rely on phone calls or manual booking methods, which can be inconvenient for patients and staff. This project aims to explore how a web-based booking system can improve accessibility, organization, and overall user experience.

HealthCare Plus is intended for:

- Patients who want an easy way to book appointments online
- Doctors who need to manage their availability
- Administrators who manage users and site content

This project will follow accessibility best practices and be responsive across desktop, tablet, and mobile devices.

2. Content Evaluation

a. Navigation

The application will use a clear and simple navigation structure to make it easy for users to find important features.

Main Navigation:

- Home
- Find a Doctor
- Appointments
- Login / Register
- Contact

Dashboard Navigation (after login):

- Patient Dashboard
- Doctor Dashboard
- Admin Panel

b. Layout and Wireframes

Wireframes will be created using Figma before development begins. These wireframes will help plan page layout, user flow, and responsive behavior.

Planned Wireframes:

- Home page
- Doctor listing page
- Doctor profile page
- Appointment booking page
- Patient dashboard
- Doctor dashboard
- Admin panel

c. Content Descriptions and Sources

Images and icons will come from free online resources such as Unsplash or open-source icon libraries. Will attach a link for Image source. Sample data will be created for testing purposes. No real medical or patient data will be used.

d. Content Types / Real-World Entities

The main content types in this project include:

- Users (patients, doctors, admins)
- Doctors
- Appointments
- Availability slots
- Pages and services

3. Functionality – User Stories

Must Have

1. Patient Account

2. As a patient, I want to create an account so that I can use the system.
3. As a patient, I want to log in so that I can access my account securely.
4. As a patient, I want to log out so that my information stays safe.
5. As a patient, I want to update my contact information so that the clinic has my latest details.
6. As a patient, I want to change my password so that my account remains secure.
7. As a patient, I want to view my profile so that I can check my information.

a. Appointment Booking

8. As a patient, I want to book an appointment online so that I do not need to call the clinic.
9. As a patient, I want to choose a doctor so that I can get the right care.
10. As a patient, I want to select a date and time so that the appointment fits my schedule.
11. As a patient, I want to see my upcoming appointments so that I can keep track of them.
12. As a patient, I want to reschedule an appointment so that I can change it if needed.
13. As a patient, I want to cancel an appointment so that the time slot is freed.
14. As a patient, I want to receive a booking confirmation so that I know it worked.
15. As an admin, I want to prevent double bookings so that appointments do not conflict.

a. Doctor

16. As an admin, I want to add doctor profiles so that patients can see available doctors.
17. As an admin, I want to update doctor information so that details stay correct.
18. As an admin, I want to remove doctor profiles so that inactive doctors are not shown.
19. As a patient, I want to browse doctor profiles so that I can choose a doctor.
20. As a patient, I want to search for doctors by specialization so that I can find the right doctor.
21. As a patient, I want to see doctor availability so that I can book an appointment.

a. Services and Departments

- 22. As an admin, I want to add services and departments so that patients know what is offered.
- 23. As an admin, I want to update service information so that it stays current.
- 24. As a patient, I want to browse services so that I can understand my options.
- 25. As a patient, I want to view department details so that I can find relevant information.
- 26. As a user, I want easy navigation so that I can find services quickly.

a. Admin and System Management

- 27. As an admin, I want to view all registered users so that I can manage accounts.
- 28. As an admin, I want to delete patient accounts so that inactive users can be removed.
- 29. As a staff member, I want to view all appointments so that schedules are organized.
- 30. As a staff member, I want to update appointment status so that records stay accurate.
- 31. As a user, I want the website to work well on all devices so that I can use it anywhere.

a. Should Have

- 32. As a user, I want form validation so that I do not submit incorrect information.
- 33. As a user, I want to see clear error messages so that I know what went wrong.
- 34. As an admin, I want different user roles so that access is controlled.
- 35. As a user, I want consistent page layouts so that the site feels easy to use.
- 36. As a user, I want the website to be accessible so that everyone can use it.

a. Nice to Have

- 37. As a patient, I want appointment reminders so that I do not forget my visit.
- 38. As a patient, I want to view my past appointments so that I can see my history.
- 39. As a patient, I want to upload documents so that doctors can review them.
- 40. As an admin, I want to see basic reports so that I can understand system usage.
- 41. As a user, I want a simple dashboard so that important information is easy to see.

1. User Login and Registration

Task	Estimated Time (hrs)
Registration form	4
Login form	3
Form validation	3
Database connection	5
Session handling	6
Error messages	5
Total	26 hrs

2. View Doctors

Task	Estimated Time (hrs)
Doctor list page	5
Search by specialization	4
Doctor profile page	7
Database queries	4
Total	20 hrs

3. Manage Doctors (Admin)

Task	Estimated Time (hrs)
Add doctor form	4
Edit doctor details	4
Delete doctor profile	6
Database operations	5
Total	19 hrs

4. Book Appointment

Task	Estimated Time (hrs)
Appointment booking form	5
Doctor availability logic	6
Store appointment in database	5
Form validation	4
Total	20 hrs

5. Manage Appointments

Task	Estimated Time (hrs)
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Appointment list page	8
Cancel appointment feature	4
Database updates	5
Total	18 hrs

6. Doctor Availability

Task	Estimated Time (hrs)
Availability management form	4
Add and remove time slots	5
Database logic	5
Total	14 hrs

7. Services and Departments

Task	Estimated Time (hrs)
Services listing page	4
Department details page	4
Database connection	3
Total	11 hrs

8. Admin and System Management

Task	Estimated Time (hrs)
User list page	6
Delete user feature	4
Database operations	4
Total	14 hrs

9. Responsive Design

Task	Estimated Time (hrs)
Mobile layout implementation	6
Tablet layout implementation	4
Responsive testing & fixes	4
Total	14 hrs

Estimated Time Summary

Priority	Estimated Time
Must Have Features	Approx 156 hours
Should Have Features	Approx 60 hours

Nice to Have Features	Approx 55 hours
Total Project Estimate	Approx 271 hours

4. Technical Specifications

a. Technology Stack

- HTML, CSS, JavaScript
- PHP with Laravel framework
- MySQL database
- Figma for wireframes
- GitHub for version control

b. Hosting Information

- Local development using XAMPP
- Deployment to shared hosting for demonstration

c. Third-Party Tools

- Laravel 10
- Bootstrap or Tailwind CSS
- Font Awesome icons +Pexels (Doctors + Patients images)

5. Data Design

a. Data Flow

Users interact with the interface, which sends requests to the server. The server processes the request, updates the database, and returns the result to the user.

b. Database Tables (Basic)

- users
- doctors
- appointments
- availability

6. Wireframes and Proposed Designs

Home Page:

Why Choose Us

Easy Scheduling

Book appointments online 24/7. Choose your preferred doctor, date, and time slot that works for you.

Expert Doctors

Access to a network of qualified and experienced doctors across multiple specialties.

Digital Records

Keep all your medical records in one secure place. Access them anytime, anywhere.

Our Services

We offer comprehensive healthcare services across multiple specialties to meet all your medical needs.

Cardiology

Comprehensive heart care including screenings, diagnostics, and treatment for cardiovascular conditions.

Pediatrics

Specialized care for infants, children, and adolescents including wellness visits and immunizations.

Orthopedics

Treatment for bone, joint, and muscle conditions including sports injuries and joint replacement.

View All Services



Login Page:

Login

Select Role

Patient

Doctor

Admin

Email

Enter email

Password

Enter Password

Login As a patient

Don't have an account? Register

Logo



Home

Find Doctors

Services

Appointment

Contact Us

Services

Contact

+1 (647) - 1234

Register Page:

Logo

Home

Find Doctor

Services

Book appointment

Contact Us

Login

Register

Register

Email

Enter email

Password

Enter Password

Phone

Enter Phone Number

Password

Enter Password

Confirm Password

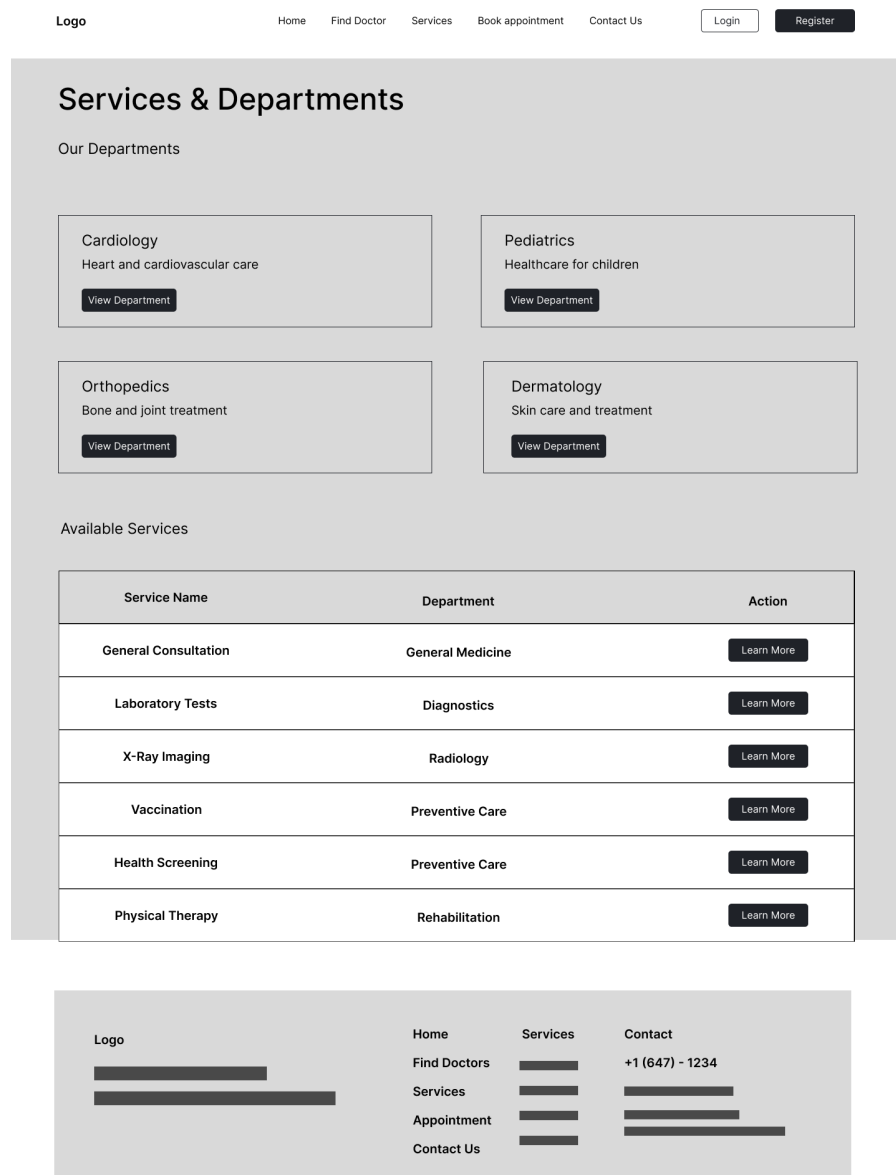
Confirm Password

Create Account

Already have an account? Login

Logo	Home	Services	Contact
	Find Doctors		+1 (647) - 1234
	Services		
	Appointment		
	Contact Us		

Service Page:



6. Development Specifications

a. Folder Structure

- public
- resources
- app
- database

b. File Naming

- Lowercase file names
- Clear and descriptive naming

c. Accessibility

- Semantic HTML
- Keyboard navigation
- Basic ARIA labels
- Color contrast considerations

d. Responsiveness

- Mobile-first design
- Works on desktop, tablet, and mobile devices

e. GitHub

- Public repository
- Weekly commits showing progress

7. Schedule

Development will follow a weekly schedule with Bernie.

8. Conclusion

HealthCare Plus is a project that demonstrates my understanding of web development concepts, including user stories, CRUD functionality, accessibility, and responsive design. This project allows me to apply the skills learned throughout the program in a practical and structured way.

